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Developing an introductory training programme for newly employed nurses in the ambulance service through participatory action research

Klara Jepsen [a,b,*,1], Jonas Wihlborg [c,2], Christoffer Ericsson [d,3], Veronica Lindström [a,e,4]

[a] Department of Neurobiology, Care Sciences and Society, Karolinska Institutet, Stockholm, Sweden
[b] Samariten Ambulans AB, Stockholm, Sweden
[c] School of health and welfare, Dalarna University, Falun, Sweden
[d] Arcada University of Applied Sciences, Academy of Business and Healthcare, Helsinki, Finland
[e] Department of Nursing, division of ambulance Service, Region Västerbotten, Umeå University, Umeå, Sweden

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ABSTRACT

Background: Newly employed nurses entering the ambulance service face clinical, organisational and cultural challenges. Although transition-related support needs have been described, less is known about how such knowledge can be translated into organisationally feasible introductory training programmes.
Aim: To explore how organisational and pedagogical challenges in introductory training for newly employed nurses were identified, interpreted, negotiated and translated into a revised programme through participatory action research (PAR).
Methods: A qualitative PAR design was conducted between 2020 and 2024 in one Swedish ambulance service organisation. Newly employed nurses, clinical supervisors, the management group (total duration 2 h and 14 min) and the research group contributed at different phases and levels of participation. The primary analytic material comprised two video-recorded management group meetings, field notes, and researcher reflections. Previously analysed interview, focus group and questionnaire findings were integrated as experiential and evaluative material. Narrative process analysis reconstructed how identified challenges were interpreted, negotiated, and translated into programme interventions, and how these were reconsidered following implementation.
Results: Four interrelated challenges were consolidated: a theory-heavy programme structure, variation in supervision, limited leadership connection and role ambiguity. Organisational interpretation and negotiation translated these into three interconnected interventions: restructuring theory and practice, preparing selected clinical supervisors and integrating peer learning during clinical placement, with leadership connection and role clarification embedded across the programme. Reflection on implementation indicated that theory–practice integration, supervision, leadership and peer learning were interdependent and required continued organisational and pedagogical support.
Conclusion: PAR provided a structured process for moving from empirical knowledge about professional transition to organisationally embedded and contextually feasible programme development. Supporting newly employed nurses requires alignment between programme design, supervision, leadership and workplace learning.
Registration: Not registered.

* Corresponding author at: Department of Neurobiology, Care Sciences and Society, Karolinska Institutet, Stockholm, Sweden.
E-mail address: klara.jepsen@ki.se (K. Jepsen).
[1] https://orcid.org/0000-0002-8070-5726.
[2] https://orcid.org/0000-0003-2171-7193.
[3] https://orcid.org/0000-0002-4086-9400.
[4] https://orcid.org/0000-0003-1386-3203.

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1. Introduction

Nurses play a central role in emergency medical services, providing safe, high-quality prehospital emergency care [1]. Newly employed nurses entering the ambulance service encounter a work environment characterised by unpredictability, rapid decision-making, and limited immediate support [2–4]. Structured introductory training programmes and supervision are therefore important for supporting professional development, confidence, and retention [5]. In Sweden, introductory training is developed locally at the organisational level, resulting in variation in programme structure, content and evaluation.

Previous research has primarily focused on newly employed nurses' transition experiences and support needs. However, less is known about how organisations translate this knowledge into organisational change and the development of introductory training programmes. This gap is important because identifying support needs does not in itself explain how organisations can translate them into feasible educational change.

Contextually adapted and evidence-informed approaches are therefore needed to integrate educational, organisational and leadership perspectives. Participatory action research (PAR) offers a collaborative framework for such development by combining practice-based knowledge with systematic inquiry through iterative cycles of planning, action, observation and reflection [6,7]. Within PAR, organisational change is understood as a collaborative process in which empirical findings, practice-based knowledge and organisational priorities are interpreted, negotiated and translated into feasible actions. This study therefore explores how identified organisational and pedagogical challenges were interpreted, negotiated and translated into a revised introductory training programme through participatory action research.

2. Background

The ambulance service is not limited to time-sensitive emergencies; it also includes independent clinical assessments, complex patient encounters, interprofessional collaboration, and responsibility for providing safe care in unpredictable and often resource-limited situations [5,8,9]. Although educational requirements vary internationally, ambulance teams commonly consist of two healthcare providers and may include nurses, first responders, Emergency Medical Technicians and/or paramedics. In several European countries, including Sweden, Finland, Belgium and the Netherlands, registered nurses constitute a core part of ambulance teams [10]. In these settings, nurses are expected to make rapid and autonomous decisions with limited immediate support, often in patients' homes, public spaces, roadside scenes or during transport [5,11,12]. These contextual characteristics make the transition into the ambulance service demanding, requiring nurses not only to adapt to a new clinical environment but also to new organisational structures, team dynamics and ways of working [5]. In Sweden, registered nurses working in ambulance services hold a bachelor's degree in nursing. In the Stockholm region, each ambulance team must include at least one specialist nurse, who has completed an additional year of university education, commonly in ambulance, anaesthesia or intensive care nursing. The second team member may be another nurse or an Emergency Medical Technician. Newly employed nurses may therefore enter the ambulance service with substantial nursing experience and advanced qualifications, while still lacking experience of autonomous assessment, decision-making and teamwork within the ambulance service context [13–15].

The transition into the ambulance service can be understood as a dynamic and situated process in which nurses navigate between previous professional experience and new contextual demands. Meleis' transitions theory provides a framework for understanding how such transitions are shaped by awareness, engagement, change and difference, time span and critical points [16]. Duchscher's concept of transition shock, although developed in relation to newly graduated nurses, is also relevant because entering a new professional context may involve uncertainty, emotional stress and temporary disruption of confidence, even among nurses with previous clinical experience [17]. For newly employed nurses in the ambulance service, this transition involves adapting to clinical demands, ad hoc team constellations, organisational expectations, role ambiguity and a workplace culture characterised by autonomy and unpredictability [2,4,5]. While transition theories explain how nurses adapt to a new professional context, they provide less guidance regarding how professional capability develops through participation in practice.

Professional development in this context involves more than acquiring clinical routines or technical skills. Dall'Alba's perspective on professional ways of being highlights that professional development also concerns how practitioners come to understand, act and become within a profession [18]. For newly employed nurses in the ambulance service, this includes developing the ability to reason, prioritise, collaborate and take responsibility in unpredictable care encounters, while negotiating professional identity, role expectations and workplace culture [5,19]. Professional development therefore depends not only on individual learning but also on organisational conditions that support supervision, reflection and workplace learning. Structured introductory training, supervision, and opportunities for reflection are therefore important for supporting professional development, confidence, and retention [5,20–22]. Peer learning (PL) is a pedagogical approach that may support such development by enabling individuals with similar levels of experience to learn with and from one another through shared practice, reflection, feedback, and mutual support [23,24].

Insufficient support during the initial phase of employment may contribute to uncertainty, emotional distress and intentions to leave, although turnover is influenced by multiple organisational, occupational and individual factors [25,26]. Introductory training should therefore not be understood as a single solution to workforce turnover. However, coherent programme structures, accessible organisational support and consistent clinical supervision may address modifiable conditions that influence professional transition, organisational belonging and decisions to remain in the organisation [22,27].

In Sweden, introductory training programmes are developed locally and vary in duration, sequencing, content, pedagogical approaches and methods of evaluation [5,22,28]. They may include theoretical education, driving training, simulation and supervised clinical placement, but the integration between these components and the preparation of clinical supervisors differ between organisations. Recent national mapping has therefore highlighted the need for more evidence-based and pedagogically coherent introductory training programmes [28].

Against this background, the present study examines how empirical knowledge of newly employed nurses' transition and support needs can be interpreted, negotiated and translated into organisational change through PAR.

3. Study aim

This study aimed to explore how organisational and pedagogical challenges in introductory training for newly employed nurses in the ambulance service were identified, interpreted and translated into a revised introductory training programme through PAR.

The study was guided by the following research questions:

1. What challenges in the existing introductory training programme were identified through the PAR process?
2. How were the identified challenges interpreted and negotiated within the PAR process?
3. How were the identified challenges translated into pedagogical and organisational programme interventions in the revised introductory training programme?

4. Methods

4.1. Design

A qualitative PAR design was used. PAR is a research strategy that addresses real-world problems through participatory and cyclical inquiry, with the dual aim of producing knowledge and supporting action [6,7]. It is widely used in healthcare to develop and improve clinical practice, particularly when practice-based knowledge, organisational dynamics and implementation processes are central [29]. Given the study's focus on the organisational change process, PAR was considered an appropriate methodological approach.

This study adopted a pragmatic and differentiated understanding of participation in PAR, in which participant groups contributed at different phases, through different forms of knowledge and with different levels of decision-making authority [7]. Newly employed nurses contributed experiential knowledge concerning transition, professional development, support needs and the implementation of the revised introductory training programme. Clinical supervisors contributed experiential and pedagogical knowledge concerning supervision and PL. The management group contributed organisational interpretation, feasibility assessment, resource negotiation and formal implementation decisions. The research group integrated these different forms of knowledge, facilitated the action-reflection process and contributed to analysis and intervention development. Participation therefore did not imply that all groups were involved equally in every phase or held equal authority over programme decisions. Accordingly, the term co-participant is used to describe individuals or groups who actively contributed to one or more phases of the PAR process, rather than to imply equal involvement in all research activities or organisational decisions.

The present study constituted the action-oriented and organisational component of a larger research project focusing on newly employed nurses' transition into the ambulance service. Findings from the preceding sub-studies [5,19,30] contributed experiential and evaluative knowledge to the PAR process by identifying transition-related, pedagogical and organisational challenges and by informing the development and subsequent reflection on the introductory training programme interventions. These datasets were not re-analysed using the original qualitative or quantitative methods in the present study. Instead, their previously analysed findings were integrated into the action-reflection process as empirical input. The primary analytic material consisted of two video-recorded management group meetings generated during the organisational negotiation and re-planning phase, supplemented by field notes and researcher reflections generated across the preceding PAR phases.

The study followed Kemmis et al.'s action research spiral, in which practice change is understood as a continuous process of planning, action, observation and reflection [6]. This cyclical structure guided the development, implementation and refinement of the introductory training programme (Fig. 1). The study was reported using relevant items from the COREQ guideline [31].

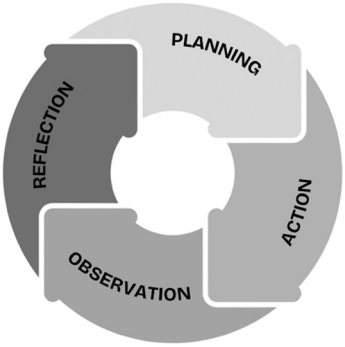


Fig. 1. The action research cycle is inspired by Kemmis, McTaggart [6].

4.2. Study setting

The study took place in Stockholm, Sweden, where two contracted organisations provided ambulance services during the study period. Samariten Ambulans, one of the two organisations contracted to the Stockholm region at the time of the study, was a privately operated ambulance service and was selected because the PAR process was embedded in an ongoing organisational development initiative within the organisation. The organisation was responsible for prehospital emergency care and operated approximately 30 ambulances daily during the study period, handling around 140,000 assignments annually throughout the Stockholm area. During the study period, approximately 22 registered nurses (RN) and 14 specialist nurses (SN) began introductory training at Samariten Ambulans AB. In this study, the term “nurses” refers to both RNs and SNs.

4.3. Participants and levels of participation

Participants contributed to the PAR process at different phases, through different forms of knowledge and with different levels of organisational influence (Table 1). The first author acted as insider researcher and process facilitator throughout the process. The research group contributed to study design, analysis, reflexive discussion and intervention development. Newly employed nurses contributed experiential and evaluative knowledge concerning transition, professional development, support needs and experiences of the revised introductory training programme. Clinical supervisors contributed experiential and pedagogical knowledge concerning supervision and PL. The management group interpreted the findings in relation to organisational conditions, assessed feasibility, negotiated priorities and made formal implementation decisions.

Across the broader research and development process, participants contributed through interviews, questionnaires, feasibility activities, management group discussions and implementation evaluations. Some individuals contributed to more than one sub-study or activity. Therefore, the figures reported in Table 1 represent contributions across datasets and should not be added together to calculate the total number of unique participants. Participation differed according to the participants' experiential, pedagogical, research and organisational roles and should not be interpreted as equal involvement in all phases or equal authority over introductory training programme decisions. Newly employed nurses and clinical supervisors primarily contributed experiential, pedagogical and evaluative knowledge, whereas the management group held formal responsibility for feasibility assessment, resource allocation and implementation decisions. At the beginning of the study, the first author had nine years of clinical experience in the ambulance service and conducted the study as an insider researcher within the participating organisation. Her clinical and organisational experience provided contextual understanding of workplace culture, clinical routines and educational practices, but also required continuous reflexivity, as described under rigour and reflexivity.

The management group consisted of the chief executive officer, managers and assistant managers responsible for the organisation's ambulance stations, a healthcare developer and a medically responsible physician. Thirteen management group members participated in the PAR process. Their ages ranged from 39 to 66 years; four were women and nine were men. Nine were specialist nurses, three were Emergency Medical Technicians and one was a physician. Eight participants had more than 21 years of ambulance service experience, four had 11–20 years and one had 5–10 years.

Newly employed nurses and clinical supervisors contributed to the PAR process through the preceding sub-studies and implementation evaluations. Eighteen newly employed nurses, including 11 RNs and

Table 1
Participant groups, contributions and levels of participation in the participatory action research process (PAR).

Participant group	Number/ contributions	Contribution and data source	PAR phase(s)	Form of participation
First author	1	Coordinated the PAR process, collected data, presented findings, facilitated management group discussions, kept field notes and researcher reflections, and contributed to analysis and intervention development	All phases	Insider research and process facilitation
Research group	**5**	Designed the study, analysed and synthesised findings, developed preliminary intervention proposals and engaged in reflexive discussion	Planning, reflection and re-planning	Research, analysis and facilitation
Newly employed nurses - transition study	18	Repeated individual interviews concerning transition experiences, support needs, role expectations and workplace culture	Planning and reflection	Experiential contribution
Nurses - professional development study	11_I 15_Q	Interviews and questionnaires concerning professionalism, professional competence, organisational support and development over time	Observation and reflection	Experiential and evaluative contribution
Newly employed nurses - PL study	10	Focus groups and individual interviews concerning experiences of PL during introductory training	Observation and reflection	Experiential and evaluative contribution
Clinical supervisors	4	Feasibility evaluation, focus group and individual interview concerning supervision, facilitation and PL	Planning, action, observation and reflection	Pedagogical and evaluative contribution
Management group	13	Video-recorded discussions, interpretation of findings, assessment of feasibility, negotiation of priorities and approval of programme interventions	Action, observation, reflection and re-planning	Organisational participation and decision-making

I: Interview Q: Questionnaire.

seven SNs, participated in repeated interviews concerning transition into the ambulance service [5]. Eleven nurses, including six RNs and five SNs, participated in longitudinal interviews concerning professionalism and professional competence, while 15 nurses completed a questionnaire after one year of employment [30]. Ten newly employed nurses, including seven RNs and three SNs, and four clinical supervisors contributed focus group and individual interview data concerning experiences of PL during introductory training [19]. These contributions provided experiential, pedagogical and evaluative knowledge that informed problem identification, intervention development and reflection on implementation.

4.4. Data collection

Data were generated and used across the iterative PAR process between 2020 and 2024, as summarised in Table 2. The process began with empirical findings from the preceding sub-studies concerning newly employed nurses' transition [5], professional development [30] and experiences of PL [19]. These previously analysed findings provided experiential and pedagogical input to identify challenges in the existing introductory training programme and served as the starting point for discussions with the management group.

In Phase 1, conducted during 2020, findings from repeated interviews with newly employed nurses were used to identify transition-related support needs, including needs for structure, reflection, role clarification and organisational belonging [5]. During the same planning phase, a small-scale PL feasibility test was conducted with four newly employed nurses and four clinical supervisors. Data from the feasibility activity included a group-based oral evaluation, field notes and researcher reflections. Together with discussions within the research group, these sources identified transition-related support needs and raised preliminary pedagogical and organisational concerns that were subsequently explored within the PAR process.

In Phase 2, during early 2021, the first author met with the management group to introduce the PAR process and the purpose of the organisational development work. Findings from the preceding sub-studies and the PL feasibility activity were presented, together with the rationale for the ongoing data generation and action research process. These meetings were not video-recorded and were used primarily to establish a shared understanding of the research process and to initiate organisational engagement with the identified transition-related and pedagogical issues.

During Phases 3 and 4, the developing process was documented through field notes and researcher reflections and discussed within the research group. These phases enabled emerging organisational responses and questions to be considered alongside the preceding empirical findings and relevant literature. The reflections were used to refine the issues requiring further organisational interpretation and to prepare the subsequent re-planning discussions with the management group. Together, Phases 1–4 progressively refined the identified challenges and possible directions for programme development. Phase 5 then constituted the recorded organisational re-planning phase in which these challenges and priorities were negotiated and translated into programme interventions.

The primary video-recorded empirical material analysed in the present study was generated during Phase 5. Two management group meetings, held on 26 October and 2 November 2021, were video-recorded to capture how the identified organisational and pedagogical challenges were interpreted, questioned, prioritised and negotiated in relation to organisational feasibility. Field notes and researcher reflections documented contextual conditions, group interactions, emerging decisions and the development of the proposed interventions. During these meetings, the identified priorities were progressively translated into intervention proposals that were discussed and revised in relation to organisational conditions. The revised programme structure, supervisor preparation and PL implementation were formally approved by the CEO in November 2021.

In Phase 6, the revised introductory training programme was implemented in spring 2022. The programme included a restructured sequence of theory and practice, preparation of selected clinical supervisors and the introduction of PL during clinical placement. In Phase 7, conducted between 2022 and 2024, implementation-related data were generated through longitudinal interviews and questionnaires concerning professional development, focus groups and individual interviews concerning experiences of PL, field notes and researcher reflections. Newly employed nurses and clinical supervisors contributed to this phase. In Phase 8, the previously analysed implementation findings were brought into reflexive discussions within the research group and compared with the challenges, priorities and expectations documented during the earlier management group discussions.

Although Phases 6–8 formed part of the overall PAR process, the interview, focus group and questionnaire datasets generated during implementation were not re-analysed in full for the present study. Instead, their previously analysed findings were used as evaluative material in the final reflection phase to examine how implementation experiences supported, challenged or extended the interpretations and introductory training programme interventions developed during Phases 2–5. The video-recorded management group meetings, field notes

Table 2
Overview of the Participatory Action Research (PAR) phases, data sources and role in the present study.

PAR phase	Time period	Purpose/activity	Material generated or used	Participants	Role in the present study
PHASE 1: PLANNING	2020	Identify transition-related support needs and explore PL as a possible pedagogical intervention	Previously analysed interview findings, PL feasibility evaluation, field notes, researcher reflections and research group discussions	Newly employed nurses, clinical supervisors, research group	Experiential and pedagogical input for problem identification
PHASE 2: ACTION	Early 2021	Introduce the PAR process, present preceding findings and establish organisational engagement	Presentation of preceding findings, PL feasibility findings, purpose of continued data generation and PAR process; field notes and researcher reflections	First author, research group, management group	Initiated organisational engagement and shared understanding
PHASE 3: OBSERVATION	2021	Capture management group discussion and decision processes	Field notes and researcher reflections	First author, management group	Contextual material informing continued problem interpretation
PHASE 4: REFLECTION	2021	Reflect on emerging issues and refine challenges requiring organisational discussion	Preceding findings, field notes, researcher reflections, literature and research group discussions	First author, research group	Primary analysis and interpretation
PHASE 5: RE-PLANNING	October–November 2021	Interpret and negotiate identified challenges and develop, revise and approve programme interventions	Two video-recorded and transcribed management group meetings, field notes, researcher reflections, intervention proposals and decisions	Management group, first author, research group	Reflection and preparation for re-planning
PHASE 6: ACTION	Spring 2022	Implement the revised introductory training programme	Revised programme structure, supervisor preparation and PL implementation	Managers, clinical supervisors, newly employed nurses, research group	Programme implementation
PHASE 7: OBSERVATION	2022–2024	Explore experiences of professional development, supervision and PL following implementation	Interviews, focus groups, questionnaires, field notes and researcher reflections	Newly employed nurses, clinical supervisors, first author	Evaluative material concerning implementation
PHASE 8: REFLECTION	2022–2024	Compare implementation experiences with earlier challenges and programme intentions	Previously analysed findings, field notes, researcher reflections and research group discussions	First author, research group; management group findings used as earlier organisational context	Final reflection and programme refinement

and researcher reflections remained the primary material for the narrative process analysis.

The data sources had different but complementary functions within the PAR process. Findings from interviews and the PL feasibility activity provided experiential and pedagogical knowledge for identifying challenges. Video-recorded management group discussions captured how these challenges were organisationally interpreted, negotiated and translated into feasible interventions. Field notes and researcher reflections documented context, turning points and the development of the process across phases. Following implementation, previously analysed interview, focus group and questionnaire findings provided evaluative knowledge concerning how the revised programme, supervision and PL were experienced. Integration occurred through the successive action-reflection cycles, in which experiential findings were presented, organisationally interpreted, translated into interventions and reconsidered in light of implementation experiences.

4.5. Analysis

The analysis was conducted as an iterative and interpretive process aligned with the PAR design [6,7]. The primary analytic focus was how organisational and pedagogical challenges were interpreted, negotiated and translated into introductory training programme interventions during the PAR process. The analysis was based on video-recorded management group meetings, field notes and researcher reflections generated during Phases 2–5 of the PAR process. The primary analysis was based on two video-recorded management group meetings from Phase 5, supplemented by field notes and researcher reflections from Phases 2–5. Findings from the preceding sub-studies [5,19,30] informed the PAR process by identifying organisational and pedagogical challenges requiring further discussion but were not re-analysed using their original qualitative or quantitative methods as part of the present study.

The material comprised two video-recorded management group meetings (total duration: 2 h and 14 min), which were transcribed by the first author using Microsoft Office transcription software. The video recordings were viewed repeatedly, while the transcripts, field notes and researcher reflections were read repeatedly to develop an overall understanding of the sequence and development of the organisational discussions and decisions. Preliminary interpretations were discussed within the research group throughout the analytical process to challenge assumptions, refine interpretations and ensure that the findings remained grounded in the empirical material [32–34].

Narrative analysis was used because the study sought to understand how organisational and pedagogical challenges were progressively interpreted, negotiated and translated into programme interventions over time [35]. Rather than analysing individual discussions as isolated events, the analysis focused on how discussions developed over time and how empirical findings from previous sub-studies were progressively interpreted, reframed and translated into feasible organisational and pedagogical interventions. Kemmis et al.'s action research spiral of planning, action, observation and reflection provided the conceptual structure for understanding this process [6].

In practical terms, the narrative analysis was conducted in four steps. First, the two management group meetings were examined chronologically. In the first meeting, the analysis identified passages in which findings from the previous sub-studies were presented and participants described or elaborated challenges in the existing introductory training programme. In the second meeting, attention was directed to how these challenges were revisited, questioned, prioritised and connected to possible changes. Second, relevant passages were compared with field notes and researcher reflections to clarify the context and sequence of the discussions. Third, the material was organised using four analytic foci: sequence, actors, turning points and translation into action (Table 3). This involved tracing who introduced or reframed an issue, where the discussion shifted from describing a problem to negotiating a response, and how the response was connected to a feasible

Table 3
Narrative analytic framework for reconstructing how organisational and pedagogical challenges were interpreted, negotiated and translated into programme interventions during the participatory action research (PAR) process.

Narrative analytic focus	What was explored in the video recordings and transcripts	Resulting interpretation	Analytic and theoretical grounding
Sequence	How discussions moved from presentation of findings to organisational response	The PAR process developed from problem identification to programme redesign	Kemmis et al.'s action research spiral [6]
Actors	How researchers and management group members contributed and reframed the discussion	Empirical findings were interpreted through experiential, research-based and organisational perspectives	Pragmatic and differentiated participation in PAR [7]
Turning points	Moments where discussions shifted from describing challenges to negotiating possible responses	Turning points concerned theory–practice integration, supervision, leadership connection, role clarification and PL.	Narrative analysis of process and change [35], interpreted in relation to professional transition and professional ways of being [17,18]
Translation into action	How identified concerns were connected to feasible programme changes	Four interrelated challenges were translated into three interconnected programme interventions.	PAR as iterative translation of knowledge into action [6,7], informed by pedagogical evidence on clinical supervision and PL [23,36]

intervention. Fourth, the first author constructed a chronological narrative linking each identified challenge to the management group's interpretation and negotiation and to the resulting programme intervention. The research group reviewed and revised this reconstruction by comparing the interpretations with the transcripts, recordings, field notes and researcher reflections.

In a subsequent integrative step, the first author systematically organised previously analysed findings from the implementation-related interview, focus group and questionnaire datasets around the four identified challenges and three introductory training programme interventions. These findings were compared to examine whether implementation experiences supported, challenged or extended the interpretations developed during the earlier phases. The original datasets were not re-coded or subjected to new thematic or quantitative analysis. Instead, the previously analysed findings were used to inform the final narrative account of implementation and reflection. The research group reviewed these comparisons to ensure that the interpretations remained consistent with the original analyses and did not exceed what the respective datasets supported.

Field notes and researcher reflections were used throughout the analysis to document contextual conditions, analytic decisions and reflexive considerations [32,33]. The final narrative reconstruction linked four interrelated organisational and pedagogical challenges to the subsequent interpretation, negotiation and development of three programme interventions: restructuring theory and practice, preparing selected clinical supervisors and integrating PL during clinical placement.

4.6. Rigour and reflexivity

Rigour was supported through methodological transparency, triangulation of data sources and investigators, and continuous reflexive discussions within the research group. Video-recorded management group meetings, field notes and researcher reflections constituted the primary empirical material for the narrative analysis, while previously analysed implementation-related findings provided evaluative material for the subsequent integrative step. Preliminary interpretations were continuously discussed within the research group to challenge assumptions, consider alternative interpretations and ensure that the narrative reconstruction remained grounded in the empirical material [33].

The first author's dual role as clinician, insider researcher and process facilitator provided contextual insight into the organisation but also required critical reflexivity. Insider–outsider positioning is a recognised dynamic in action research, particularly when participation, power and organisational embeddedness may shape the inquiry [7,32,34]. To address this, the first author kept a reflective journal, and regular discussions with co-researchers were used to scrutinise assumptions, interpretations and the influence of the researcher's dual role on the analytical process [32]. The research group's complementary experience in ambulance service, nursing, education and qualitative methodology contributed to critical discussions and alternative perspectives throughout the analytical process. Particular attention was paid to how the first author's insider position might influence the interpretation and narrative reconstruction of the PAR process.

Participation was voluntary, and informed consent was obtained from all participants. Confidentiality and anonymity were protected through data de-identification and careful reporting. The study adhered to the ethical principles of the Declaration of Helsinki [37] and was approved by the Stockholm Ethical Review Board (no. 2019–06138).

5. Results

The narrative analysis reconstructed how organisational and pedagogical challenges were identified, interpreted, negotiated and translated into programme interventions during the PAR process. The results are presented in four analytic phases addressing problem identification (RQ1), organisational interpretation and negotiation (RQ2), translation into programme interventions (RQ3), and subsequent implementation and reflection. These phases should not be understood as a direct one-to-one representation of Kemmis et al.'s action research cycle [6].

5.1. From a fragmented introduction to shared problem understanding

The video-recorded management group meetings in Phase 5 marked a more focused stage of the organisational interpretation process that had been initiated during earlier discussions with the management group. Previously analysed findings concerning newly employed nurses' transition and professional development, together with the PL feasibility activity, field notes, and researcher reflections, were presented as the basis for further discussion. The management group subsequently elaborated and reframed these challenges in relation to organisational conditions and the existing introductory training programme structure. Through these discussions, a shared understanding emerged that the existing introductory training did not provide a sufficiently coherent learning pathway from theoretical preparation to situated practice.

As the discussions progressed, four interrelated organisational and pedagogical challenges were consolidated as central to the management group's shared interpretation of the existing introductory training programme. First, findings from newly employed nurses indicated that the existing programme was experienced as theory-heavy, with extensive information delivered early and limited opportunities to connect theoretical content with the contextual and practical aspects of ambulance service practice. Nurses described a need for earlier familiarisation with equipment, the ambulance environment, simulation-based learning and practical problem-solving.

Second, interview and evaluation findings from newly employed

nurses indicated variation in supervision. Supervisory experiences varied in opportunities for reflection, feedback, gradual responsibility and educational dialogue. Experiences from the PL feasibility activity further indicated that supervisors required clear expectations and preparation for a facilitative pedagogical role. Together, these findings suggested that learning conditions depended substantially on the individual supervisor rather than on a shared pedagogical structure.

Third, findings from newly employed nurses showed a limited connection to leadership during the introductory period. Some nurses were uncertain about who their immediate manager was, particularly when placed in the staff pool or when contact with the assigned station was limited. This contributed to uncertainty regarding where organisational support could be sought and weakened early organisational belonging.

Fourth, interview findings identified ambiguity concerning RN and SN roles and the distribution of responsibility within ambulance teams. Newly employed nurses described uncertainty about what was expected of them and how formal responsibilities related to everyday team practices. The management group discussions further highlighted how informal workplace norms and expectations from senior colleagues could reinforce this ambiguity.

Together, these discussions shifted the understanding of transition from primarily individual adaptation towards an organisational responsibility for workplace learning, supervision, leadership contact and role expectations. The four challenges thus formed the analytical bridge between preceding empirical findings and subsequent organisational negotiation. This shared understanding marked a turning point from problem identification towards the development of feasible introductory training programme interventions.

5.2. Negotiating organisational and pedagogical priorities

The four identified challenges were carried forward into the next phase of the PAR process, where the management group considered their organisational significance, feasibility and prioritisation within existing operational conditions. The theory-heavy structure was discussed in relation to how learning activities were sequenced and connected across the introductory training programme. Management group members reflected on earlier introductory formats that had included more practical exercises, such as equipment handling and training in challenging environments. One participant stated: “It was more practical training… how valuable it is to really get to know your equipment” (Participant 1). As the discussion developed, the challenge was reframed from a need for additional content to a need for stronger integration between theoretical education and practice-based learning. A shared understanding emerged that practical experience should be introduced earlier and linked more closely to subsequent theoretical education.

Variation in supervision was discussed as a source of unequal learning conditions. The management group considered how the selection and preparedness of clinical supervisors could shape newly employed nurses' early experiences. As one participant stated: “The wrong clinical supervisor can give a negative start” (Participant 2). The discussion reframed supervision from an informal arrangement dependent on individual supervisors to an educational and organisational responsibility. A shared understanding developed that clinical supervisors should be selected and prepared for their pedagogical role.

The limited connection to leadership was considered in relation to organisational belonging and clarity of support. The management group discussed how responsibility for the introduction had become partially detached from station managers and how this could contribute to uncertainty among newly employed nurses. One participant described “a feeling of not knowing which manager they actually have” (Participant 3). Through the discussion, the group developed a shared understanding that managerial contact needed to be established earlier and more clearly during the introductory process.

Role ambiguity was interpreted in relation to workplace culture, team expectations and professional socialisation. The management group discussed how informal norms within ambulance teams could challenge formal descriptions of RN and SN roles. Senior staff were described as important culture carriers: “The culture in the organisation is passed on to the new [employees]. This can be both positive and negative” (Participant 2). The discussion led to a shared understanding that newly employed nurses needed structured opportunities to reflect on responsibility, role expectations and professional practice, rather than relying solely on informal socialisation within ambulance teams. The discussions also revealed tensions between pedagogical priorities and operational feasibility, particularly regarding workload and supervisory capacity. These concerns became part of the subsequent intervention development.

Together, these discussions transformed the four identified challenges into four organisational and pedagogical priorities: stronger integration of theory and practice, more consistent and pedagogically prepared supervision, clearer leadership connection and structured opportunities for reflection and role clarification. This shared prioritisation provided the basis for the next phase of the PAR process, in which the priorities were translated into concrete programme interventions within the revised introductory training programme.

5.3. Translating priorities into programme interventions

The four organisational and pedagogical priorities identified in the preceding phase were carried forward into the introductory training programme development process. Rather than producing one intervention per priority, the research and management groups combined them within a feasible programme structure into three interconnected interventions: restructuring theory and practice, preparing selected clinical supervisors and integrating PL. Leadership connection and role clarification were embedded across these interventions.

The priority of strengthening theory–practice integration was translated into a revised programme structure. The previous eight-week introductory training programme comprised four weeks of theoretical education, one week of driving training and three weeks of clinical practice. The revised nine-week introductory training programme reorganised these learning activities to deliberately alternate between theoretical education, driving training and clinical practice. As illustrated in Fig. 2, two initial weeks of theoretical education were followed by one week of driving training and two weeks of PL-based clinical placement. Newly employed nurses then returned for two further weeks of theoretical education before completing two weeks of individual clinical practice with an experienced ambulance team. Practical and simulation-based activities were also introduced earlier, allowing experience from practice to inform subsequent theoretical learning and strengthening the intended theory–practice integration.

Rather than becoming a separate intervention, leadership connection was incorporated across the revised programme through earlier managerial contact. Management representatives participated in the initial theoretical component to introduce themselves, clarify organisational functions and establish early contact with newly employed nurses. This embedded response addressed the previously identified uncertainty concerning managerial belonging within the introductory training programme structure.

More consistent supervision was addressed through a more deliberate selection and preparation of clinical supervisors. Supervisors were selected based on interest in education and previous supervisory or educational experience and completed a half-day preparation session before PL was implemented. The preparation addressed PL principles,



Fig. 2. Structure of the revised introductory training programme.

facilitation of reflection, learner support and the supervisor's responsibility for maintaining patient safety while enabling newly employed nurses to take an active role in patient encounters. This intervention represented a shift from relying primarily on individual supervisory experience towards establishing clearer organisational expectations for the pedagogical role.

The priorities concerning reflection, role clarification and shared learning were translated into PL during clinical placement. Newly employed nurses completed two weeks of practice in peer teams, preferably pairing an RN with an SN, or two RNs when an SN was unavailable. The peers participated jointly in patient encounters, alternated leading roles and were encouraged to verbalise assessments, discuss decisions and reflect on their respective roles and actions. Clinical supervisors remained responsible for patient safety and clinical guidance but were expected to facilitate learning through questions, feedback and reflection rather than primarily directing practice. PL therefore provided a structured learning space in which role expectations, clinical reasoning and professional responsibility could be explored within situated practice. Together, the three interventions formed a coordinated organisational response: introductory training programme sequencing addressed theory–practice integration, supervisor preparation addressed pedagogical consistency, and PL supported reflection and role clarification, while leadership contact was embedded across the revised introductory training programme.

Translation into action also required negotiation of organisational feasibility. Management group members raised concerns regarding workload, the availability of suitable clinical supervisors and the capacity to implement PL within everyday operations. These concerns did not lead to rejection of the proposed interventions but influenced how the introductory training programme was organised and highlighted the need to balance pedagogical ambitions with staffing and operational conditions. Organisational negotiation therefore formed an integral part of intervention development rather than occurring only after the introductory training programme had been designed. Following these discussions, the CEO formally approved the revised programme structure, supervisor preparation and PL implementation in November 2021. This decision marked the transition from programme development to organisational action. The revised introductory training programme was subsequently implemented in spring 2022, enabling the interventions to be examined through the observation and reflection phases of the PAR process.

5.4. Refining the programme through reflection on implementation

Previously analysed findings from interviews, focus groups and questionnaires, together with field notes and researcher reflections, were used as reflection points to examine how the revised introductory training programme addressed earlier challenges and where further refinement was needed, rather than as a separate outcome evaluation.

Reflection on the revised programme suggested that introducing practical learning earlier and reconnecting it with subsequent theoretical education was perceived as creating a more coherent learning pathway. However, participants continued to request more simulation-based exercises, stronger theoretical preparation and structured feedback on patient encounters, indicating a need for further pedagogical development.

Reflection on supervisor preparation and leadership visibility indicated that early managerial contact, structured supervision and opportunities for peer dialogue supported security and confidence, but these benefits remained vulnerable to variation between supervisors, stations and clinical placements. Supervisor preparation and leadership involvement were therefore understood as ongoing organisational responsibilities rather than components of a one-off introductory training programme.

PL supported shared reflection, role clarification and mutual learning during clinical placement, but depended on prepared supervisors and a workplace culture that enabled dialogue and active participation. It was therefore understood as supporting the programme's pedagogical aims only when embedded within supportive supervisory and organisational conditions.

Taken together, implementation-related reflections indicated that theory–practice integration, supervision, leadership connection and PL were interdependent rather than separate programme components. The revised programme was therefore understood as an ongoing organisational and pedagogical development process, with reflection informing subsequent re-planning and further refinement.

6. Discussion

The main contribution of this study is to demonstrate how empirical knowledge of newly employed nurses' transition into the ambulance service can be translated into organisationally embedded interventions within introductory training programmes through a PAR process. Rather than treating transition as an individual adaptation process, the present narrative analysis showed how empirical findings, practice-based knowledge and organisational priorities were progressively interpreted and negotiated during the development of a revised introductory training programme. Supporting transition was therefore understood not solely as an educational challenge but as an organisational responsibility requiring alignment among pedagogical design, leadership and workplace learning. In this way, the study extends previous research by moving beyond identifying support needs to demonstrate how such knowledge can be operationalised through organisational programme development. This addresses recent national findings showing substantial variation in Swedish ambulance service introductory training programmes and a need for more evidence-based and pedagogically coherent approaches [28].

A central finding of the narrative analysis was that the existing introductory training was organisationally interpreted as failing to provide a sufficiently coherent pathway from theoretical preparation to situated practice within ambulance teams. This gap between theory and practice is particularly important in the ambulance service, where newly employed nurses are expected to reason, prioritise and act in unpredictable environments with limited immediate support [5,9]. The revised introductory training programme addressed this priority through programme redesign, introducing clinical practice earlier and more deliberately alternating theory, driving training, and practice. From Dall'Alba's perspective on professional ways of being, this restructuring can be understood as a pedagogical attempt to support professional development as more than the acquisition of routines or technical skills; it was intended to support nurses' developing ways of understanding, acting and becoming within ambulance service practice [18]. The restructuring was therefore not merely a practical adjustment but a pedagogical response to the situated nature of professional learning, illustrating how PAR can support introductory training programme structures that more closely align workplace learning with professional practice.

Findings from the preceding sub-studies identified supervision as an important condition for transition and professional development, including variation in opportunities for reflection, feedback and educational dialogue. The present narrative analysis showed how these experiential findings were subsequently reframed by the management group as an educational and organisational responsibility rather than solely an issue of individual supervisory practice. This aligns with previous research showing that clinical supervision can support nurses' confidence, self-efficacy, communication, and professional development [36,38]. In contrast, insufficient support during transition may contribute to insecurity and stress when entering a new professional role [17]. Preparing selected clinical supervisors represented the translation of this priority into a structured programme intervention to strengthen the pedagogical quality and consistency of workplace learning. The present study extends this knowledge by demonstrating how supervision

can be established as an organisational responsibility through the development of a structured introductory training programme within a PAR process.

Leadership operated at two interconnected levels in the PAR process. Implementation-related findings indicated that, for newly employed nurses, early managerial contact was experienced as contributing to clarity, support and organisational belonging. At the organisational level, the present narrative analysis showed that formal leadership provided the authority required to translate participatory deliberation into programme change. Previous research has linked accessible and supportive leadership with staff well-being, job satisfaction and organisational commitment [39,40]. The present analysis extends this perspective by showing that leadership also functioned as a mechanism through which collaboratively developed priorities could be authorised, resourced and implemented. Thus, while PAR emphasises participation and collaborative knowledge development, organisational change also depended on formal decision-making authority.

The present narrative analysis showed that workplace culture and role ambiguity were closely linked to supervision and leadership. The management group recognised that newly employed nurses not only learn clinical routines but are also socialised into local norms, expectations and ways of working. From Dall'Alba's [39] perspective, this can be understood as part of professional becoming, whereby nurses develop ways of being and acting within ambulance service practice rather than merely acquiring knowledge and skills. Workplace culture may therefore both support and constrain professional development. Previous research similarly describes professional socialisation as involving the internalisation and negotiation of professional norms, values and role expectations [40,41]. The present analysis extends this understanding by showing how concerns related to role ambiguity and workplace culture were translated into structured reflection and programme design rather than being left to informal socialisation alone.

PL represented the pedagogical translation of the identified need for shared reflection and role clarification during clinical placement. The implementation-related findings indicated that PL was experienced as supporting a safe learning space and mutual support, while reducing the sense of isolation associated with entering a new role. This aligns with previous research showing that PL can strengthen collaboration, reflection and collective learning in healthcare education and workplace introduction [23,24,42,43]. In terms of professional acting, PL may provide opportunities for newly employed nurses to enact emerging understanding into professional action through shared reasoning, decision-making and reflection in clinical situations [39]. Taken together, the narrative analysis and implementation-related findings indicated that PL could not be understood as a stand-alone educational strategy. Its implementation was described as depending on prepared supervisors, leadership support and a workplace culture that valued dialogue, reflection and shared learning. This illustrates how pedagogical innovations introduced through PAR depend on organisational conditions that enable their integration into everyday practice.

Beyond the specific context of one Swedish ambulance service organisation, the present analysis contributes to broader questions of how healthcare organisations can structure workplace learning during professional transition. Similar tensions concerning theory–practice integration, supervision, organisational support and role development have been described in other ambulance service and emergency healthcare contexts [2,4,9]. Rather than suggesting that the specific interventions in the introductory training programme are directly transferable, the present study illustrates a participatory process through which empirical knowledge, practice-based experience and organisational feasibility were brought together in the development of the introductory training programme. Such an approach may be relevant in other complex, team-based healthcare settings where introductory training must address professional socialisation, role clarity and organisational belonging alongside clinical learning.

Overall, the narrative analysis and subsequent reflection on implementation suggest that introductory training should be understood as an organisational process that integrates workplace learning, professional transition, and programme development. A coherent introductory training programme therefore appears to require alignment between theory and situated practice, prepared supervisors, visible leadership and structured opportunities for reflection. Taken together, the narrative analysis and implementation-related findings indicated that these components were interdependent rather than discrete educational interventions and required continued organisational and pedagogical support. In this sense, the PAR process provided a means of moving from empirical knowledge about professional transition towards organisationally embedded and contextually feasible programme development. Although developed within the ambulance service, this approach may also inform the design of introductory training and workplace learning in other complex, team-based healthcare settings.

7. Strengths and limitations

Credibility was strengthened by the PAR design, which enabled experiential knowledge from newly employed nurses and clinical supervisors to be brought into an organisational process in which identified challenges were progressively connected to organisational programme development. The use of video-recorded management group meetings, field notes, and researcher reflections enabled source triangulation and facilitated tracing of how identified challenges were interpreted and connected to programme interventions.

Explicitly distinguishing the levels and forms of participation increased methodological transparency. Newly employed nurses and clinical supervisors primarily contributed experiential and pedagogical knowledge, whereas the management group contributed organisational interpretation, feasibility assessment and implementation decisions. Participation and decision-making power were therefore not equal, and the management group's formal authority may have influenced which interventions were considered feasible and prioritised. Co-creation in this study should consequently not be understood as equal participation across all phases. Experienced ambulance service perspectives were included through clinical supervisors and management group members, many of whom had substantial ambulance service experience. However, senior clinically practising nurses without supervisory or managerial roles were not recruited as a separate participant group. Their inclusion might have provided additional perspectives on workplace culture, informal learning and the feasibility of the revised programme.

The first author's insider position was both a strength and a limitation. Clinical experience and organisational embeddedness provided contextual understanding of routines, workplace culture and educational practices and facilitated continuous engagement throughout the iterative PAR process. However, this position may also have influenced data generation, interpretation and interactions with participants. Reflexive field notes, researcher reflections and discussions within the research group were used to examine assumptions and strengthen credibility [33].

PAR also created methodological challenges related to the volume, heterogeneity and fragmentation of material generated across phases and for different purposes. The primary narrative analysis was based on two management group meetings supplemented by field notes and researcher reflections, while previously analysed implementation-related findings were incorporated in a subsequent integrative step. This created a risk that the narrative reconstruction could privilege the perspectives of organisational decision-makers or selectively connect events occurring at different times. To strengthen dependability and confirmability, the function of each data source was specified, the analytic sequence and decisions were documented, and interpretations were repeatedly compared with the empirical material and discussed within the research group [34]. Nevertheless, the differentiated levels of participation and the first author's dual role may have shaped which issues were raised, how they were negotiated and how the process was

reconstructed.

The study was conducted within a single Swedish ambulance service organisation, which limits transferability. Rather than assuming that the specific programme interventions can be transferred directly, the description of the organisational context, participant roles, data sources, analytic process and intervention development enables readers to assess the applicability of the findings to their own settings [34]. The process may be relevant to other complex, team-based healthcare settings where workplace learning, professional transition and organisational development intersect. Further research is needed to examine longer-term effects on professional development, professional competence, retention, patient safety and organisational sustainability.

8. Conclusions

This study explored how organisational and pedagogical challenges in introductory training for newly employed nurses in the ambulance service were identified, interpreted, negotiated and translated into programme interventions through PAR. The findings show that transition into the ambulance service is shaped not only by individual adaptation but also by how organisations structure workplace learning, supervision, leadership contact and role expectations.

Through shared problem understanding and organisational negotiation, four interrelated challenges were translated into a revised programme comprising a restructured sequence of theory and practice, preparation of selected clinical supervisors and PL during clinical placement, with leadership connection and role clarification embedded across these components. Reflection on implementation further indicated that these components were interdependent and required continued organisational and pedagogical support. Overall, the study demonstrates how PAR can provide a structured process for moving from empirical knowledge about professional transition to organisationally embedded and contextually feasible programme development. Although developed within one ambulance service organisation, this process may be relevant to other complex, team-based healthcare settings.

9. Implications for practice

Ambulance service organisations should view introductory training as an organisational process integrating workplace learning, professional transition and programme development, rather than primarily as information transfer. Development of introductory training may benefit from participatory approaches combining empirical knowledge, practice-based experience and organisational decision-making. Programmes should provide a coherent pathway between theoretical preparation and situated clinical practice, with early opportunities for practical learning, simulation, reflection and feedback.

Clinical supervisors should be deliberately selected and prepared for their pedagogical role to promote consistent learning conditions. Leadership should be visible early in the introductory period to clarify organisational structures, role expectations and available support, and remain engaged beyond the initial programme period. These components should be considered interdependent parts of the organisational conditions supporting workplace learning rather than separate educational activities.

PL may provide a structured approach to shared learning, reflection and role clarification during clinical placement when supported by prepared supervisors and a workplace culture enabling dialogue and active participation. Its implementation should therefore be considered part of a broader organisational strategy for introductory training rather than an isolated pedagogical intervention.

CRediT authorship contribution statement

Klara Jepsen: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Jonas Wihlborg:** Writing – review & editing, Validation, Supervision. **Christoffer Ericsson:** Validation, Writing – review & editing. **Veronica Lindström:** Writing – review & editing, Validation, Supervision, Data curation, Conceptualization.

Ethics approval and consent to participate

The study was approved by the Stockholm Ethical Review Board, Sweden (Ref. no.: 2019–06138). The study followed the ethical principles of the Declaration of Helsinki, ensuring anonymity, integrity and confidentiality. Participants received written and verbal information about the study. Participation was voluntary, with the right to withdraw, and informed consent was obtained before participation.

Disclosure statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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